

Absence of Cubic Nonlinearities in a Curvature Functional for 3D Navier–Stokes and a Conditional Small-Data Regularity Criterion

Luiz Puodzius*

Independent Researcher, Oakland Park, Florida

June 2026

Abstract

We introduce a curvature-based functional $\Phi(C)$ for the 3D Navier–Stokes equations. A symbolic expansion ($\approx 1.8\text{M}$ characters) provides **computational evidence** that $\frac{d}{dt}\Phi(C)$ contains **no pure cubic terms** of the form $(\partial u)^3$. Assuming a conjectured quartic bound consistent with this structure, we prove a **conditional small-data global regularity theorem**.

The remaining obstacle is a half-derivative gap, formulated as the **Half-Derivative Bridge** conjecture. Amplitude-scaling constraints are analyzed to clarify what forms such an inequality could take.

Why this matters. The curvature functional $\Phi(C)$ isolates the nonlinear structure of Navier–Stokes in a way that removes the cubic vortex-stretching obstruction at the differential level. This makes it a promising tool for conditional regularity, restricted geometries, and the search for new coercive inequalities.

1 Introduction

For a divergence-free velocity field u on the periodic domain $\mathbb{T}^3$, the vorticity $\omega = \nabla \times u$ satisfies

$$\partial_t \omega + (u \cdot \nabla) \omega = (\omega \cdot \nabla) u + \nu \Delta \omega. \quad (1)$$

*ORCID: 0009-0009-3616-4115. SFITSuite@proton.me

The enstrophy evolution contains the vortex-stretching term

$$\int_{\mathbb{T}^3} \omega \cdot (\omega \cdot \nabla) u \, dx, \quad (2)$$

which is cubic in first derivatives and cannot be controlled by dissipation alone. This is the classical obstruction to closing energy estimates for 3D Navier–Stokes.

We introduce a curvature functional whose evolution *eliminates this cubic term entirely* at the differential level, as evidenced by symbolic computation. This structural property enables a conditional small-data regularity theorem and motivates the precise open problem formulated in Section 7.

2 The Curvature Functional

Definition 1 (Vorticity tensor).

$$T_{ij} = \partial_i u_j - \partial_j u_i.$$

Definition 2 (Curvature).

$$C_{\mu\nu i} = \partial_\mu T_{\nu i} - \partial_\nu T_{\mu i} + T_{\mu j} \partial_j T_{\nu i} - T_{\nu j} \partial_j T_{\mu i}. \quad (3)$$

Definition 3 (Curvature functional).

$$\Phi(C) = \frac{1}{2} \int_{\mathbb{T}^3} \sum_{\mu, \nu, i=1}^3 C_{\mu\nu i}^2 \, dx. \quad (4)$$

Remark 4. The curvature $C_{\mu\nu i}$ mixes second derivatives of u (through the linear part $\partial_\mu T_{\nu i} - \partial_\nu T_{\mu i}$) with quadratic first-derivative interactions (through the Lie-bracket terms $T_{\mu j} \partial_j T_{\nu i}$). The functional $\Phi(C)$ therefore controls a combination of $\|\nabla^2 u\|_{L^2}$ and $\|\nabla u\|_{L^4}^2$.

3 Computational Evidence for Structural Cancellation

Observation 5 (Symbolic computation). Using SymPy, we expanded $\frac{d}{dt}\Phi(C)$ in terms of u and its derivatives via equation (1). The resulting expression contains approximately 1.8 million characters. Coefficient hashing detected **no monomials** of the form

$$(\partial_{a_1} u_{i_1})(\partial_{a_2} u_{i_2})(\partial_{a_3} u_{i_3}),$$

for any indices $i_1, i_2, i_3 \in \{1, 2, 3\}$ and directions a_1, a_2, a_3 .

Remark 6 (Caveat). Observation 5 is **computational evidence**, not an analytic proof. A complete analytic derivation of the absence of cubic terms remains open. The symbolic code and coefficient-hashing logs are available at the repository listed in the Appendix.

Observation 7 (Fourier-mode check). For a single Fourier mode $u(x) = A \sin(k \cdot x)e$ on $\mathbb{T}^3$, direct computation gives

$$\frac{d}{dt}\Phi(C) = -2\nu k^4 A^2 + O(A^4), \quad (5)$$

with **no cubic term** $O(A^3)$. This is consistent with Observation 5.

Remark 8. The Fourier-mode calculation (5) provides an explicit, independently verifiable instance of the absence of cubic terms. At leading order in amplitude A , the functional decays dissipatively with rate $2\nu k^4$.

4 Conditional Differential Inequality

Conjecture 9 (Quartic bound). *There exists a constant $K > 0$ (depending only on the domain) such that*

$$\left| \frac{d}{dt}\Phi(C) + 2\nu \|\nabla C\|_{L^2}^2 \right| \leq K \Phi(C)^2 \quad (6)$$

for all smooth divergence-free $u \in C^\infty(\mathbb{T}^3)$.

Remark 10 (Justification). The symbolic expansion of $\frac{d}{dt}\Phi(C)$ contains no cubic terms. The remaining nonlinearities are quartic — of the form $|\omega|^2 |\nabla \omega|^2$, $|\omega|^4$, and similar — which are dimensionally consistent with the scaling $\Phi(C)^2$. Conjecture 9 is the precise analytic formulation of this structure.

5 Elliptic Coercivity

Lemma 11 (Elliptic coercivity of the linear part). *The linearization of the map $u \mapsto C$ at $u = 0$,*

$$C_{\mu\nu i}^{(0)} = \partial_\mu T_{\nu i} - \partial_\nu T_{\mu i},$$

is an elliptic operator of order 2. On $\mathbb{T}^3$:

$$\|C^{(0)}\|_{L^2}^2 \geq \lambda_0 \|\nabla^2 u\|_{L^2}^2 \quad (7)$$

for some constant $\lambda_0 > 0$.

Proof sketch. In Fourier space, $C_{\mu\nu i}^{(0)}$ involves second-order differences of the Fourier coefficients of u . The symbol is non-degenerate for $k \neq 0$, yielding the coercivity estimate by

Plancherel's theorem and the Poincaré inequality on $\mathbb{T}^3$. $\square$

Lemma 12 (Nonlinear perturbation). *Write $C = C^{(0)} + C^{(1)}$, where $C^{(1)}$ contains the Lie-bracket terms $T_{\mu j} \partial_j T_{\nu i}$. There exists $\delta > 0$ such that if $\Phi(C) < \delta$, then*

$$\|\nabla C\|_{L^2}^2 \geq \lambda \Phi(C), \quad \lambda = \lambda_0/2. \quad (8)$$

Proof sketch. The nonlinear part $C^{(1)}$ is quadratic in first derivatives of u . For small $\Phi(C)$, the H^2 norm of u is small (by Lemma 11), so $C^{(1)}$ is dominated by $C^{(0)}$ in L^2 . The coercivity of $C^{(0)}$ then yields the estimate with $\lambda = \lambda_0/2$. $\square$

6 Conditional Small-Data Global Regularity

Theorem 13 (Conditional small-data regularity). *Assume Conjecture 9. Define the critical threshold*

$$\epsilon_c = \frac{\nu\lambda}{2K}.$$

If the initial data satisfies

$$\Phi(C(0)) < \min(\epsilon_c, \delta),$$

where δ is the coercivity threshold of Lemma 12, then:

1. $\Phi(C(t))$ decays exponentially:

$$\Phi(C(t)) \leq \Phi(C(0)) e^{-\frac{\nu\lambda}{2}t};$$

2. *the solution u is global and smooth.*

Proof. From Conjecture 9 and Lemma 12:

$$\frac{d}{dt}\Phi(C) \leq -\nu\lambda \Phi(C) + K \Phi(C)^2. \quad (9)$$

If $\Phi(C(0)) < \epsilon_c = \nu\lambda/(2K)$, a bootstrap argument shows $K \Phi(C(t)) \leq \nu\lambda/2$ for all $t \geq 0$. Substituting into (9):

$$\frac{d}{dt}\Phi(C) \leq -\frac{\nu\lambda}{2} \Phi(C).$$

Grönwall's lemma yields the exponential decay in part (1).

For part (2): by Lemma 11, smallness of $\Phi(C)$ implies smallness of $\|\nabla^2 u\|_{L^2}$, hence smallness of $\|u_0\|_{H^2}$. By the classical Fujita–Kato small-data theorem [1], small H^2 (equivalently $\dot{H}^{1/2}$) initial data yields a unique global smooth solution. $\square$

Remark 14. Theorem 13 does not provide a new small-data theorem beyond Fujita–Kato. Its significance is structural: it identifies a new functional $\Phi(C)$ whose smallness implies

the Fujita–Kato condition, and whose evolution is governed by a differential inequality free of cubic terms.

7 The Half-Derivative Bridge

To extend Theorem 13 to arbitrary data, one needs to control $\|\nabla^2 \omega\|_{L^2}$ — required by the Beale–Kato–Majda criterion [3] — from $\Phi(C)$.

Conjecture 15 (Half-Derivative Bridge). *There exist a constant $C_0 > 0$ and a continuous non-decreasing function $F : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that*

$$\|\nabla^2 \omega\|_{L^2}^2 \leq F(\Phi(C)) + C_0 \|\omega\|_{L^2}^2 \quad (10)$$

for all smooth divergence-free u .

Remark 16 (Amplitude-scaling constraint). A linear bound $F(x) = Cx$ is impossible by scaling analysis:

- the left-hand side of (10) scales as A^2 for a solution of amplitude A ,
- the nonlinear part of $\Phi(C)$ scales as A^4 .

Therefore any valid F must satisfy

$$F(x) \gtrsim x^{1/2} \quad \text{for large } x. \quad (11)$$

This constraint rules out linear F and narrows the search space for the inequality.

Corollary 17. *If Conjecture 15 holds, then Conjecture 9 can be upgraded to an unconditional statement, and Theorem 13 extends to arbitrary smooth initial data: global regularity holds for all $T < \infty$.*

8 Summary

Claim	Status
Absence of cubic terms in $\frac{d}{dt}\Phi(C)$	Computational evidence (1.8M chars, zero cubic matches)
Quartic bound (Conjecture 9)	Supported by evidence; analytic proof open
Small-data regularity (Theorem 13)	Conditional on Conjecture 9
Half-Derivative Bridge (Conjecture 15)	Open; scaling requires $F(x) \gtrsim x^{1/2}$

This framework provides a precise, testable conjecture, a conditional theorem, and a structural discovery (absence of cubic terms) that motivates further analytic investigation.

Appendix: Computational Details

Symbolic code and coefficient-hashing logs are available at:

<https://github.com/Dr-Loo/3D-Navier-Stokes-regularity>

Key output: zero cubic monomials detected among $\approx 1.8\text{M}$ characters of the SymPy expansion of $\frac{d}{dt}\Phi(C)$.

References

- [1] H. Fujita and T. Kato, “On the Navier–Stokes initial value problem I,” *Archive for Rational Mechanics and Analysis* **16**, 269–315 (1964).
- [2] T. Kato, “Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions,” *Mathematische Zeitschrift* **187**, 471–480 (1984).
- [3] J. T. Beale, T. Kato, and A. Majda, “Remarks on the breakdown of smooth solutions for the 3-D Euler equations,” *Communications in Mathematical Physics* **94**, 61–66 (1984).
- [4] L. Puodzius, “SFIT: A Slow–Fast Design Pattern for Structured Algorithmic Domains,” *Zenodo* (2026). DOI: 10.5281/zenodo.20596022